



{Computer Programming 1}

Project Report

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1. Introduction

The "Smart Parking Management System" is a console-based software application developed in the C programming language. Its primary purpose is to modernize and automate the traditional manual process of managing vehicle parking. Instead of relying on paper logs, this system allows operators to dynamically manage parking capacity, track vehicle entries and exits, calculate parking fees based on vehicle type, and generate automatic receipts. The system simulates a real-world environment using dynamic memory allocation to ensure efficiency and scalability.

2. Project Objectives

The main objectives of this project are:

- **Dynamic Memory Management:** To demonstrate the ability to handle memory manually using malloc and free, allowing the system to adapt to different parking lot sizes at runtime.
- **Data Organization:** To utilize advanced data structures like structs, enums, and bit-fields to store complex vehicle data efficiently.
- **String Manipulation:** To apply standard string library functions for validating inputs (e.g., license plate formats) and processing text.
- **Algorithmic Logic:** To implement search algorithms (Linear Search) for finding vehicles and Recursion for calculating financial data.
- **Data Persistence:** To use File I/O operations to save transaction records (receipts) permanently.

3. Code Discussion & Mechanism

The program operates on a continuous loop that serves as the main control panel. Upon starting, the system asks the user to define the parking capacity. It then allocates the exact amount of memory required using malloc.

The core workflow is as follows:

1. **Visualization:** The system renders a grid view of the parking slots, using color codes (Green for Free, Red for Occupied) to give the user immediate visual feedback.
2. **Parking (Input):** When a user selects to park a car, the system searches for the first available slot. It validates the license plate (checking for length and format) before saving the data.
3. **Removing (Output):** When a car exits, the system searches for the license plate using string comparison. It calculates the fee based on the vehicle type (Sedan, SUV, or Truck) and the time spent.
4. **Recursive Reporting:** A recursive function traverses the array to calculate potential revenue without using standard loops, demonstrating algorithmic skill.
5. **File Logging:** Upon exit, a receipt is appended to a text file (receipt.txt).

4. List of Functions

This section lists both the **User-Defined Functions** (created for this project) and the **Standard Library Functions** (from C libraries) used to build the logic.

A. User-Defined Functions

Function Name	Description
main	The entry point of the program. It handles memory allocation, the main menu loop, and calls other functions based on user choice.
visualizeParking	Renders a visual grid of the parking lot on the screen, showing occupied slots in Red and empty ones in Green.
parkCar	Handles the logic of adding a vehicle. It finds an empty slot, validates input strings, and stores vehicle details.
removeCar	Handles vehicle exit. It searches for a plate number, calculates the fee, saves the receipt to a file, and frees the slot.
calculateRevenueRecursive	A recursive function that calculates the total estimated revenue from currently parked cars.
saveReceipt	Opens a file named receipt.txt and appends the transaction details (Name, Plate, Cost) for permanent storage.
generateWelcomeMessage	A helper function that uses string concatenation to create a personalized welcome message for the driver.
clearScreen	Clears the console window to keep the user interface clean (supports both Windows and Linux/Mac).
printHeader	Prints the stylized title and version of the program at the top of the screen.

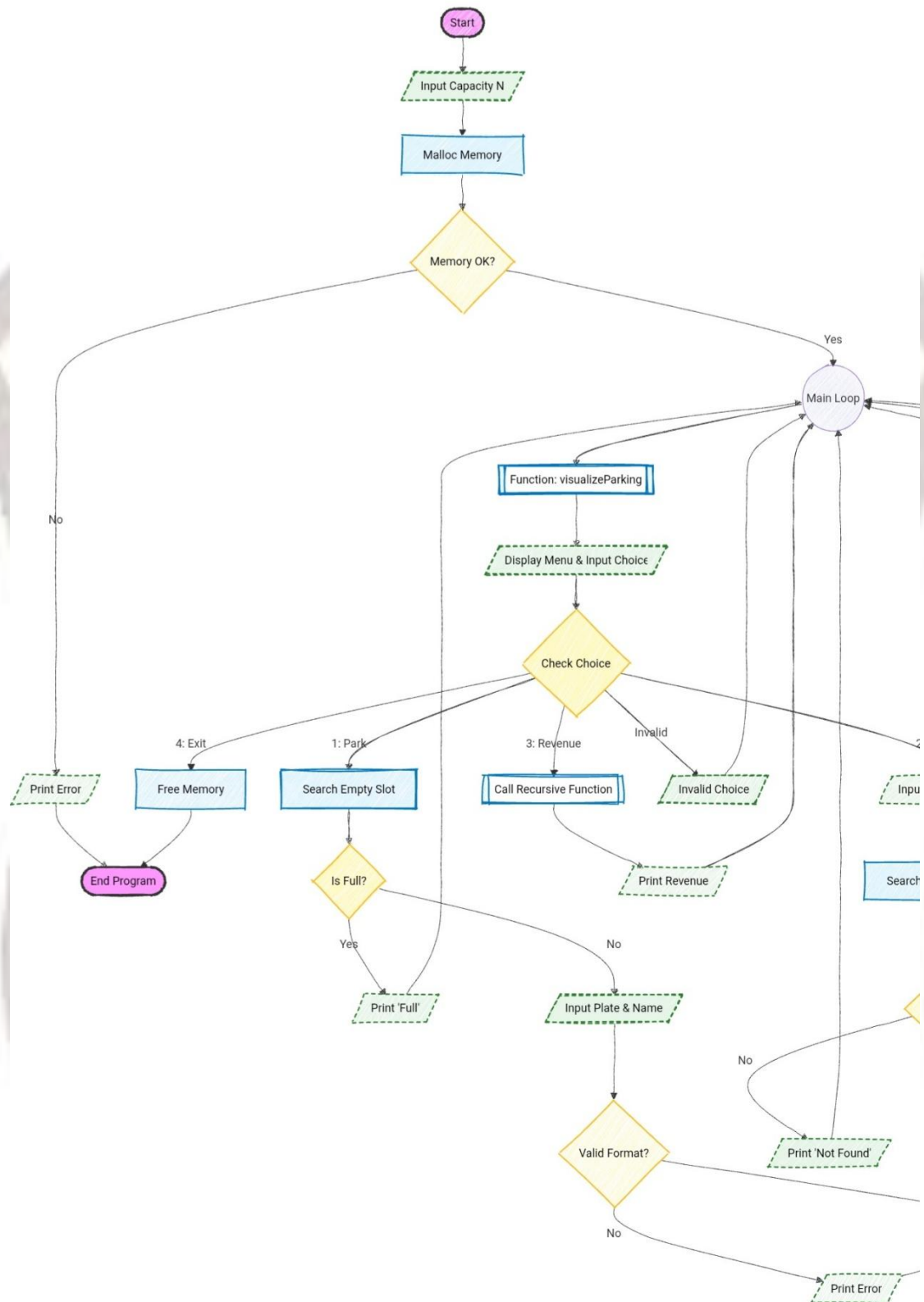
B. Standard Library Functions (Used from C Libraries)

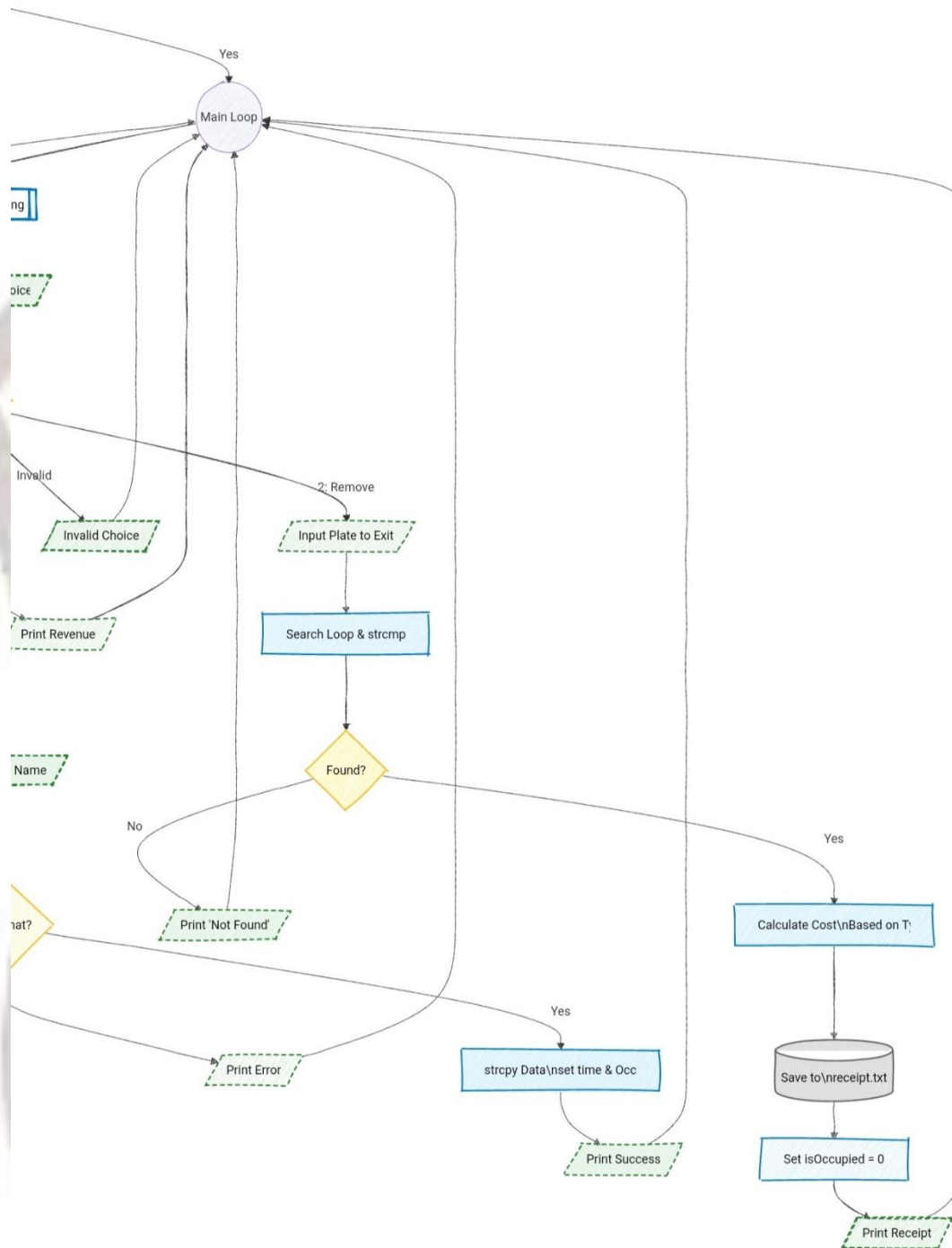
Function Name	Library	Description
printf	<stdio.h>	Outputs formatted text and numbers to the console screen.
scanf	<stdio.h>	Reads formatted input from the user (keyboard).
Function Name	Library	Description
getchar	<stdio.h>	Reads a single character (used here to pause the screen and clear input buffers).
fopen	<stdio.h>	Opens a file (used for the receipt system).
fprintf	<stdio.h>	Writes formatted text into a file.
fclose	<stdio.h>	Closes an open file to save changes.
malloc	<stdlib.h>	Allocates a block of memory dynamically during runtime.
free	<stdlib.h>	Deallocates (releases) the memory back to the system.
system	<stdlib.h>	Executes system commands (used for cls or clear).
strlen	<string.h>	Calculates the length of a string (used to validate plate length).
strcpy	<string.h>	Copies a string from one variable to another.
strchr	<string.h>	Searches for a specific character in a string (used to check for '-' in plates).
strcmp	<string.h>	Compares two strings (used to find the car by plate number).
strcat	<string.h>	Concatenates (joins) two strings together.
time	<time.h>	Returns the current calendar time (used to timestamp entry).

5. Flowchart

Description of the Flowchart:

1. **Start:** User launches the program.
2. **Input:** User enters Capacity.
3. **Process:** Program allocates memory using malloc.
4. **Decision Node (Loop):** Display Menu.
 - **If Choice 1 (Park):** Search for empty slot -> (Full? Print Error) / (Empty? Input Data -> Validation -> Save).
 - **If Choice 2 (Remove):** Search for Plate -> (Found? Calculate Cost -> Save File -> Clear Slot) / (Not Found? Print Error).
 - **If Choice 3 (Revenue):** Call Recursive Function -> Print Total.
 - **If Choice 4 (Exit):** Break Loop.
5. **End:** Free memory -> Terminate.

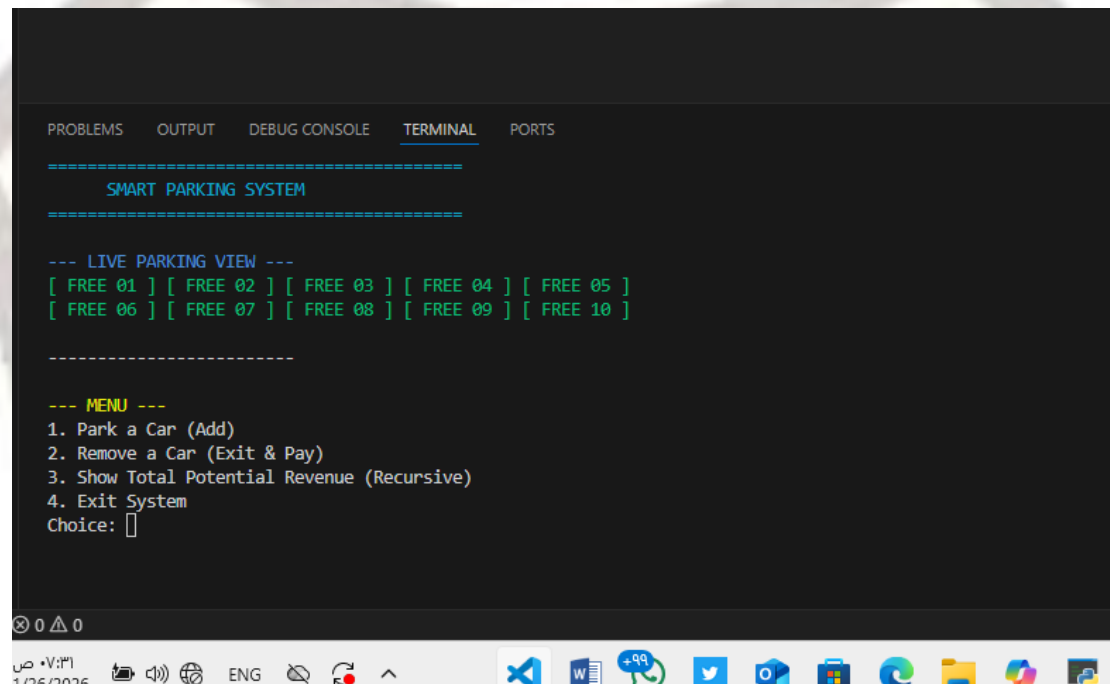




6. Input/Output Screenshots (Simulation)

Below are representations of the program's runtime interface.

A. Initialization & Visual View



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PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

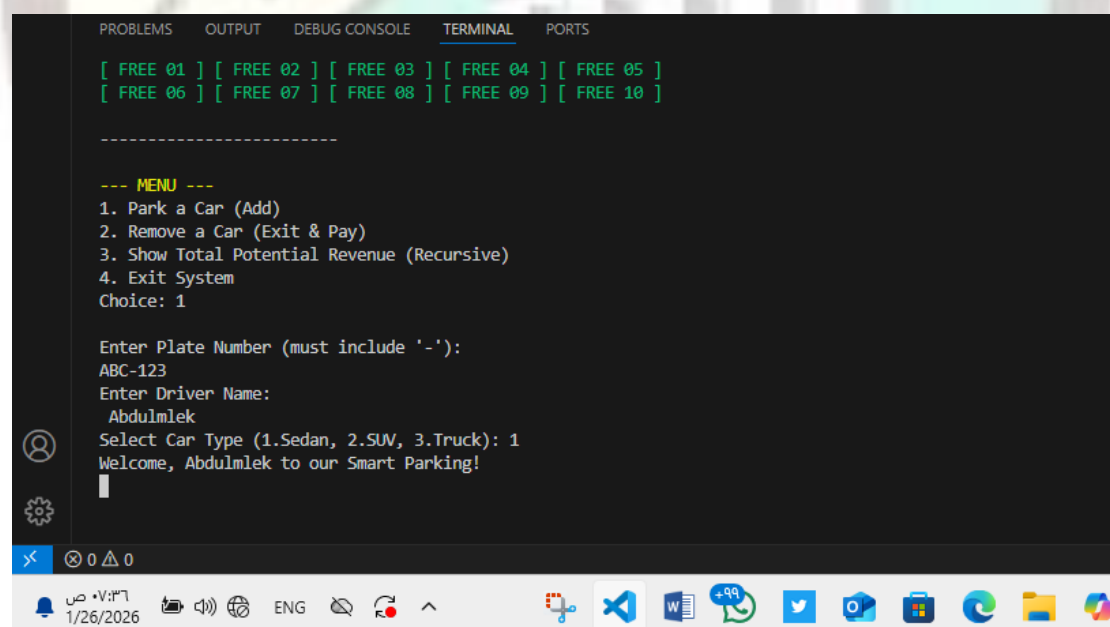
=====
SMART PARKING SYSTEM
=====

--- LIVE PARKING VIEW ---
[ FREE 01 ] [ FREE 02 ] [ FREE 03 ] [ FREE 04 ] [ FREE 05 ]
[ FREE 06 ] [ FREE 07 ] [ FREE 08 ] [ FREE 09 ] [ FREE 10 ]

-----

--- MENU ---
1. Park a Car (Add)
2. Remove a Car (Exit & Pay)
3. Show Total Potential Revenue (Recursive)
4. Exit System
Choice: 
```

B. Parking a Car (With Validation)



```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

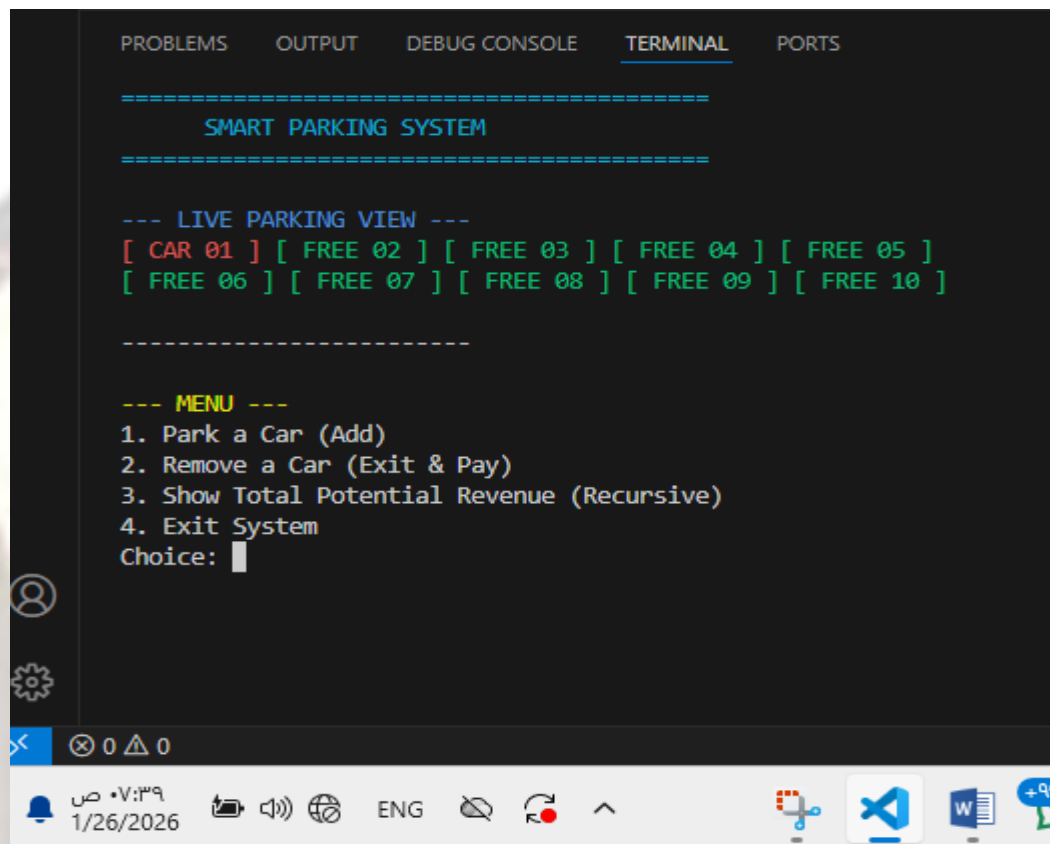
[ FREE 01 ] [ FREE 02 ] [ FREE 03 ] [ FREE 04 ] [ FREE 05 ]
[ FREE 06 ] [ FREE 07 ] [ FREE 08 ] [ FREE 09 ] [ FREE 10 ]

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--- MENU ---
1. Park a Car (Add)
2. Remove a Car (Exit & Pay)
3. Show Total Potential Revenue (Recursive)
4. Exit System
Choice: 1

Enter Plate Number (must include '-'):
ABC-123
Enter Driver Name:
Abdulmlek
Select Car Type (1.Sedan, 2.SUV, 3.Truck): 1
Welcome, Abdulmlek to our Smart Parking!
|
```

C. Menu Updated View



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PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

=====
SMART PARKING SYSTEM
=====

--- LIVE PARKING VIEW ---
[ CAR 01 ] [ FREE 02 ] [ FREE 03 ] [ FREE 04 ] [ FREE 05 ]
[ FREE 06 ] [ FREE 07 ] [ FREE 08 ] [ FREE 09 ] [ FREE 10 ]

-----

--- MENU ---
1. Park a Car (Add)
2. Remove a Car (Exit & Pay)
3. Show Total Potential Revenue (Recursive)
4. Exit System
Choice: |
```

D. Removing a Car & Receipt Generation

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

=====
SMART PARKING SYSTEM
=====

--- LIVE PARKING VIEW ---
[ CAR 01 ] [ FREE 02 ] [ FREE 03 ] [ FREE 04 ] [ FREE 05 ]
[ FREE 06 ] [ FREE 07 ] [ FREE 08 ] [ FREE 09 ] [ FREE 10 ]

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--- MENU ---
1. Park a Car (Add)
2. Remove a Car (Exit & Pay)
3. Show Total Potential Revenue (Recursive)
4. Exit System
Choice: 2
Enter Plate Number to Exit: ABC-123
Car Found! Total Cost: $10.00
Car Removed & Receipt Saved.
█
```

E. Recursive Revenue Calculation

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PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

=====
SMART PARKING SYSTEM
=====

--- LIVE PARKING VIEW ---
[ CAR 01 ] [ CAR 02 ] [ CAR 03 ] [ FREE 04 ] [ FREE 05 ]
[ FREE 06 ] [ FREE 07 ] [ FREE 08 ] [ FREE 09 ] [ FREE 10 ]

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--- MENU ---
1. Park a Car (Add)
2. Remove a Car (Exit & Pay)
3. Show Total Potential Revenue (Recursive)
4. Exit System
Choice: 3

Estimated Revenue from current cars: $60.00
Press Enter to continue...█
```

